

UBAT TERKAWAL/CONTROLLED MEDICINE

JAUHKAN DARIPADA CAPAIAN KANAK-KANAK

KEEP OUT OF REACH OF CHILDREN

FOR ANIMAL USE ONLY

Methadyne[™] **Injection 10mg/ml**

For use only by a registered veterinarian.

For analgesia and restraint in horses, and analgesia and anaesthetic premedication in dogs and cats.

METHADYNE 10 % INJECTION

(METHADONE HYDROCHLORIDE 10mg/ml)

Active Substance

Methadone Hydrochloride 10
mg/ml (Equivalent to 8.9 mg/ml
methadone)

Product Description

A clear colorless injectable solution packed in a 20ml amber glass vial type 1, with chlorobutyl stopper and non-reusable aluminum flip off cap.

Pharmacodynamics

Methadone is structurally unrelated to other opium-derived analgesics and exists as a racemic mixture. Each enantiomer has a separate mode of action; the d-isomer non-competitively antagonizes the NMDA receptor and inhibits norepinephrine reuptake; the l-isomer is a μ -opioid receptor agonist.

There are two subtypes μ_1 and μ_2 . The analgesic effects of methadone are believed to be mediated by both the μ_1 and μ_2 subtypes, whereas the μ_2 subtype appears to mediate respiratory depression and inhibition of gastrointestinal motility. The μ_1 subtype produces supraspinal analgesia and the μ_2 receptors produce spinal analgesia.

Methadone has the ability to produce profound analgesia. It can also be used for premedication and it can assist in the production of sedation in combination with tranquilizers or sedatives. The duration of effect may vary from 1.5 to 6.5 hours.

Opioids produce a dose-dependent respiratory depression. Very high doses may result in convulsions.

Pharmacokinetics

In dogs, methadone is absorbed very rapidly (T_{max} 5-15 minutes) following intramuscular injection of 0.3 to 0.5 mg/kg. T_{max} tends to be later at the higher dose levels indicating that an increase in dose tends to prolong the absorption phase. The rate and extent of systemic exposure of dogs to methadone appears to be characterized by dose-independent (linear) kinetics following intramuscular administration. The bioavailability is high and ranges between 65.4 and 100%, with a mean estimate of 90%. Following subcutaneous administration of 0.4 mg/kg, methadone is absorbed slower (T_{max} 15-140 minutes) and bioavailability is $79 \pm 22\%$.

In dogs, the volume of distribution at steady state (V_{ss}) was 4.84 and 6.11 l/kg in males and females respectively. The terminal half-life is in the range 0.9 to 2.2 hours following intramuscular administration, and is independent of dose and sex. The terminal half-life may be slightly longer following intravenous administration. The terminal half-life ranges from 6.4 to 15 hours following subcutaneous administration. Total plasma clearance (CL) of methadone following intravenous administration is high 2.92 to 3.56 l/h/kg or ca 70% to 85% of the cardiac plasma output in dogs (4.18 l/h/kg).

In cats, methadone is also rapidly absorbed following intramuscular injection (peak values occur at 20 minutes), however, when the product is administered inadvertently subcutaneously (or in another poorly vascularized area) absorption will be slower.

The terminal half-life is in the range of 6 to 15 hours. Clearance is medium to low with a mean (sd) value of 9.06 (3.3) ml/kg/min.

Methadone is extensively protein bound (60% to 90%). Opioids are lipophilic and weak bases. These physiochemical properties favor intracellular accumulation. Consequently, opioids have a large volume of distribution, which greatly exceeds total body water. A small amount (3% to 4% in the dog) of the administered dose is excreted unchanged in the urine; the remainder is metabolized in the liver and subsequently excreted.

Indication

For analgesia and restraint in horses, and analgesia and anaesthetic premedication in dogs and cats.

Recommended Dosage

HORSES

Analgesia: 0.22 mg/kg (10 ml/450 kg) bodyweight by IM/ SC injection

Restraint: 0.11 mg/kg (5 ml/450 kg) bodyweight combined with 0.055 mg/kg bodyweight acetylpromazine (2.5 ml of 10 mg/ml acetylpromazine per 450 kg) by IV Injection

DOGS

Analgesia: 0.11 - 0.55 mg/kg (0.11 - 0.55ml/ 10 kg) bodyweight by IM/SC Injection

Premedication: 1.1 mg/kg (1.1 ml/10 kg) bodyweight by SC Injection

CATS

Analgesia: 0.2 - 0.6 mg/kg (0.2 - 0.6 ml/10 kg) bodyweight by SC Injection

Premedication: 0.2 - 0.6 mg/kg (0.2 - 0.6 ml/10 kg) bodyweight by SC Injection

Repeat analgesia: 0.2 - 0.6 mg/kg (0.2-0.6 ml/10 kg) bodyweight by SC Injection at 3 timepoints with 5-hour intervals

Contraindications

This product is contraindicated for use in animals during parturition

This product is contraindicated for use in animals with respiratory depression.

This product is contraindicated for use in pregnant, breeding or lactating cats as safe use of this product in these animals has not been established.

Warnings and Precautions

Use with caution in animals with hypothyroidism, adrenocortical insufficiency, impaired renal and hepatic function and shock. Published information supports the safe and effective use of methadone in cats older than 6 months of age.

Interactions with Other Medicaments

Methadone can potentiate the effects of analgesics, central nervous system inhibitors and substances that cause respiratory depression. Concomitant or subsequent use of the veterinary medicinal product with buprenorphine may lead to lack of efficacy

When used in combination with other products, lower dosages can be used. For safe use with other Veterinary medicinal products, reference must be made to the relevant product literature

Pregnancy and Lactation

Methadone diffuses across the placenta.

Studies in laboratory animals have shown adverse effects on reproduction.

The safety of the product during pregnancy and lactation has not been assessed in the target species. The use of the product is not recommended during pregnancy or lactation.

Adverse Effects

In very common cases (more than 1 in 10 animals displaying adverse reaction(s) during the course of one treatment), the following reactions have been observed after administration of the product:

Cats: Respiratory depression may be seen. Mild excitatory reactions have been observed: lip licking, vocalization, urination, defecation, mydriasis, hyperthermia and diarrhea. Hyperalgesia has been reported. All reactions were transient.

Dogs: Respiratory depression and bradycardia may be seen. Mild reactions have been observed: panting, lip licking, salivation, vocalization irregular breathing, hypothermia, fixed stare and body tremors. Occasional urination and defecation can be seen within the first hour post dose. All reactions were transient.

Symptoms and Treatment of Overdose

Cats: In case of overdoses (>2 mg/kg) the following signs can be observed:

increased salivation, excitation, hind leg paralysis and loss of righting reflex. Seizures, convulsion and hypoxia were also recorded in some cats. A dose of 4 mg/kg could be fatal in cats. Respiratory depression has been described.

Dogs: Respiratory depression has been described. Methadone can be antagonized by naloxone. Naloxone should be given to effect. A starting dose of 0.1 mg/kg intravenously is recommended

Incompatibilities

The product is incompatible with injection fluids containing meloxicam, or any other non-aqueous solution.

This veterinary medicinal product can be mixed in the same syringe with aqueous solutions for injection containing acepromazine as maleate and medetomidine and dexmedetomidine as hydrochlorides.

Syringes with these mixtures should be used as soon as practicable. Any unused mixed solution remaining in the syringe should be disposed appropriately.

In the absence of further compatibility studies, this veterinary medicinal product must not be mixed with other veterinary medicinal products.

Withdrawal Period

MEAT WITHHOLDING PERIOD (HORSES): DO NOT USE in horses that may be used for human consumption

Storage Conditions

Store below 30° C.

Protect from light.

Use the contents within 28 days of first broaching of the vial.

Discard unused portion

Dosage Forms and Packaging Available

Injectable solution available in packaging of 20 ml.

REG NO: MAL XXXXXXXXX

PRODUCT REGISTRATION HOLDER

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